

WEEKLY PROGRESS REPORT

BEHAVIORAL RISK SCORING FRAMEWORK & REAL-TIME C2 MITIGATION

1. CLINICAL WORK OVERVIEW

This week marks the migration of the Intrusion Prevention System (IPS) from isolated binary thresholds to an advanced Behavioral Risk Scoring Framework (0-100 scale). This dynamic architecture integrates Lexical, Temporal, and Volumetric analysis to detect sophisticated evasion tactics (Micro-slicing, Low & Slow). The system was validated against live C2 infrastructure using the dnscat2 framework, proving its capability to enforce kernel-level isolation while eliminating memory-tracking bugs.

2. CORE SYSTEM ARCHITECTURE UPDATES (METHODOLOGY)

2.1 Dynamic Risk Scoring Matrix (0-100 Scale)

To avoid threshold-bypassing attacks, every source IP is evaluated dynamically:

- Lexical Layer (Max 30 pts): AI model classification (20 pts) + L7 DPI Heuristic Override safety net (10 pts).
- Temporal/Burst Frequency Layer (Max 35 pts): Tracks unique subdomain mutations within a 20s sliding window (12 pts per unique hit, capped at 35).
- Volumetric Accumulation Layer (Max 35 pts): Tracks total data exfiltrated per root domain via a linear ratio (Current Bytes / 1000B * 35, capped at 35).

2.2 Mitigation Tier Mapping

- GREEN Tier (<40): Normal internet activity; system logs silently.
- YELLOW Tier (40-74): Suspicious anomalous footprint; activates enhanced monitoring for evasion.
- RED Tier (>=75): Critical threat verified; triggers active response via kernel-level iptables isolation.

2.3 Memory Leak & State Reset Fix (Tuple Key Resolution)

A silent architectural bug was resolved in the active response module. The tracking system mapped exfiltration volume using a compound tuple key: (src_ip, root_domain). The unblock function was refactored using list comprehension to scan and erase all tracking sub-keys bound to the isolated host upon penalty expiration, preventing an infinite-blocking loop bug.

3. SIMULATION & VERIFICATION RESULTS

3.1 Live Traffic Testing with dnscat2

The framework was subjected to dynamic C2 interactive traffic:

- Initial Handshake & Discovery: Captured cleanly under the GREEN/YELLOW boundary (Score: 20-56/100).
- High-Volume Commands Execution (e.g., cat /etc/passwd, file download): Triggered immediate accumulation into the RED Tier (Score: 85/100).

3.2 Active Response Countermeasures

- Attacker Perspective (Kali Linux): The interactive command channel suffered immediate freeze. Attempts to execute commands threw "Window not found" errors, and keep-alive "Ping!" packets failed to retrieve a response.
- Victim Perspective (Debian Client): After 20 unsuccessful retransmission attempts through the blocked gateway, the client terminated the tunnel automatically with a FATAL session exit status.
- Gateway Perspective (Ubuntu IPS): Successfully executed automated iptables rule insertion and flushed states safely upon timeout.